International Institute of Information Technology - Hyderabad

EC5.203 - Communication Theory

Final Exam

April 28, 2026

- Number of questions: 7; Total points: 40; Time Limit: 3 hours
- **Read all instructions and questions carefully.** No step marks will be given if you try to answer a question that is different from the one asked.
- Use of a calculator is permitted.
- This is a closed-book exam.
- Clearly write your assumptions or use of any properties at each step.
- Even if the final answer is correct, only *partial marks* will be given if the approach or methodology is incorrect, not presented clearly, or not legible.
- Even if the final answer is incorrect or incomplete, *partial marks* may be given if the approach or methodology is presented clearly.
- Do not write anything on the question paper! There will be a penalty for marking any answers (including MCQ) on the question paper.
- Follow notations/conventions from class, if not explicitly stated.

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1. State whether the following statements are True or False. Give a reason for your statement in 2-3 sentences. No step marking: both the statement and reason must be right. **(10 points)**

   (a) A mixer cannot be implemented using non-linearity.

   (b) An envelope detector can be used as one of the building blocks for demodulating an FM signal.

   (c) PM can be used to generate an FM signal.

   (d) QPSK is not possible for a single physical baseband channel, while it is possible in a single physical passband channel.

   (e) Although noise PSD is considered flat, noise power seen by the receiver is finite.

   (f) Applying the Gram-Schmidt procedure on a set of signals, we get one unique set of basis signals.

   (g) For 3-ary constellation containing symbols at (0,1), (0,-1), and (2,0), $P_e = P_{e|i}$ for all $i = 0, 1, 2$.

   (h) For ML, the average probability of error P_e does not depend on the prior probabilities.

   (i) For orthogonal M-FSK with fixed symbol energy, the probability of error decreases as M increases.

   (j) 1024-QAM is generally suitable for traditional satellite links with nonlinear amplification.

2. Draw the block diagram of a digital communication system. Explain each block in 2-3 lines? (**5 points**)

3. Prove the following for the conventional AM (**5 points**)

   (a) The power efficiency for conventional AM is given by $\eta_{\text{AM}} = \frac{a_{\text{mod}}^2\,\overline{m_n^2}}{1+a_{\text{mod}}^2\,\overline{m_n^2}}$.

   (b) Further prove that $\eta_{\text{AM}} \leq 50\%$ in general.

   (c) Prove that $\eta_{\text{AM}} \leq 1/3$ for sinusoidal message signal $m(t) = A_m \cos(2\pi f_m t)$.

4. Answer the following for Angle Modulation **(5 points)**

   (a) Prove that the angle-modulated waveforms have infinite bandwidth theoretically. As a special case, derive the narrowband FM expression and its bandwidth.

   (b) Find the bandwidth of the FM signal corresponding to a sinusoidal message $m(t) = 0.5\,\sin 2\pi f_m t$.

5. Consider that the received signal $y(t) = s(t) + n(t)$, where $s(t)$ is a deterministic signal, corresponding to a specific choice of transmitted symbols and $n(t)$ is zero-mean white noise with PSD $S_n(f) = \frac{N_0}{2}$. Considering real-valued signals, answer the following questions (**5 points**)

   (a) Show that the SNR at the output of the correlator is given by

   $$\text{SNR} = \frac{|< s, g >|^2}{\frac{N_0}{2}||g||^2},$$

   where $g(t)$ is an arbitrary reference signal.

   (b) Find $g(t)$ which will maximize the output SNR? Also, find the maximum SNR.

6. Consider the binary hypothesis testing problem:

   $$\begin{aligned} H_0 &: Y \sim \mathcal{U}(0,1), \\ H_1 &: Y \sim \mathcal{U}(\theta, \theta+1), \end{aligned} \quad \text{where } 0 < \theta < 1,$$

   with prior probabilities $P(H_0) = p$ and $P(H_1) = 1 - p$. Answer the following questions in this context (**5 points**)

(a) Write the probability density functions under both hypotheses and derive the likelihood ratio $\Lambda(Y)$.

(b) Using the likelihood ratio test, identify the decision regions.

(c) Derive the MAP decision rule (in terms of comparing Y to a threshold) and express the threshold in terms of the prior probabilities.

(d) Obtain expressions for the conditional probabilities of error and the average probability of error.

7. For the 8-QAM constellation with points (1,1), (1,0), (1,-1), (0,-1), (-1,-1), (-1,0), (-1,1), and (0,1), solve the following (**5 points**)

   (a) Draw the constellation and carefully sketch the ML decision regions

   (b) Based on the decision regions in part (a), find the union bound, intelligent union bound and the nearest neighbor approximations for the average symbol error probability in terms of E_b/N_0. Assume equally likely symbols and $E_b = E_s/\log_2 M$.

   (c) Suggest a modification to the constellation to improve error performance without increasing average energy. Justify your design qualitatively.